Real-World Applications & Global Change Projections

Modeling the distribution of biodiversity under global climate change

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Future climate change projections

Unravelling biogeographic patterns and future changes

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RESEARCH ARTICLE OPEN ACCESS

Global Cold-Water Coral Biodiversity Redistribution Under Projected Climate Change

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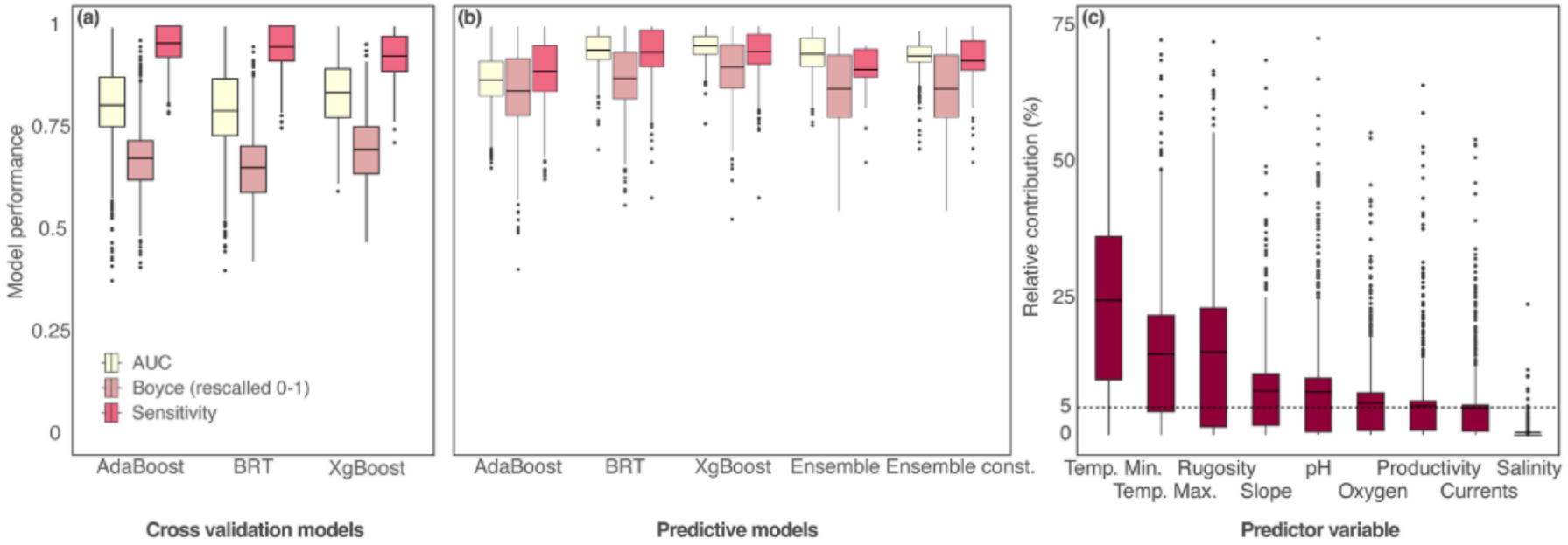
Keywords: biodiversity | bioregionalization | climate change | cold-water corals | range shifts | species distribution modelling | species richness

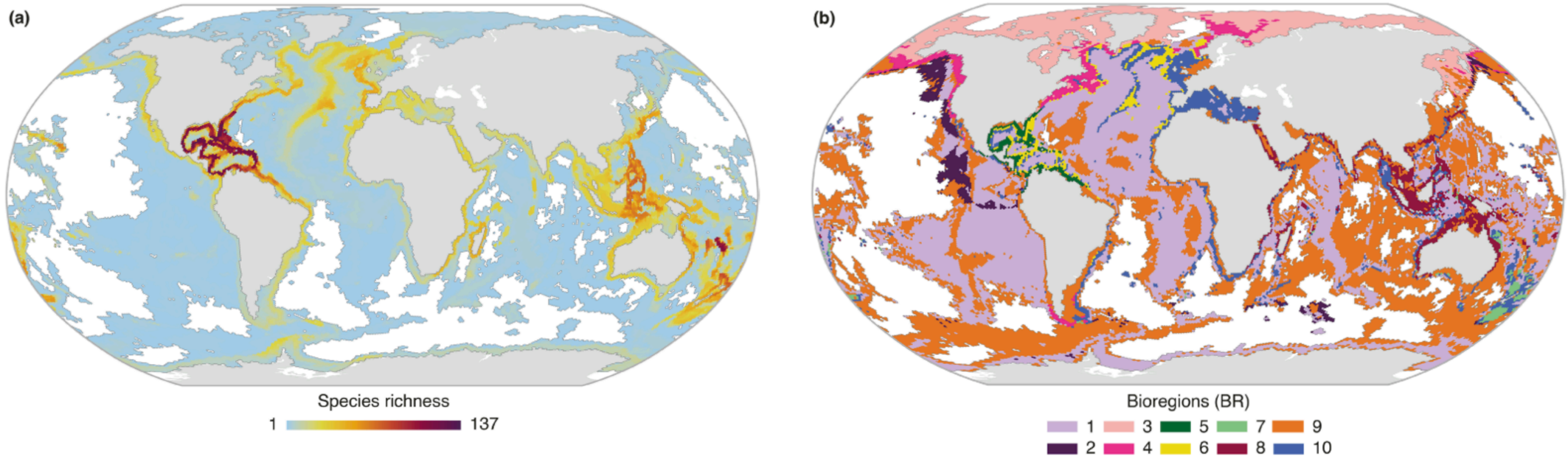
ABSTRACT

Rationale: Cold-water corals (CWCs) are **key ecosystem-structuring species** across the world's oceans, **yet their global distribution, diversity patterns, and vulnerability to climate change remain poorly understood**.

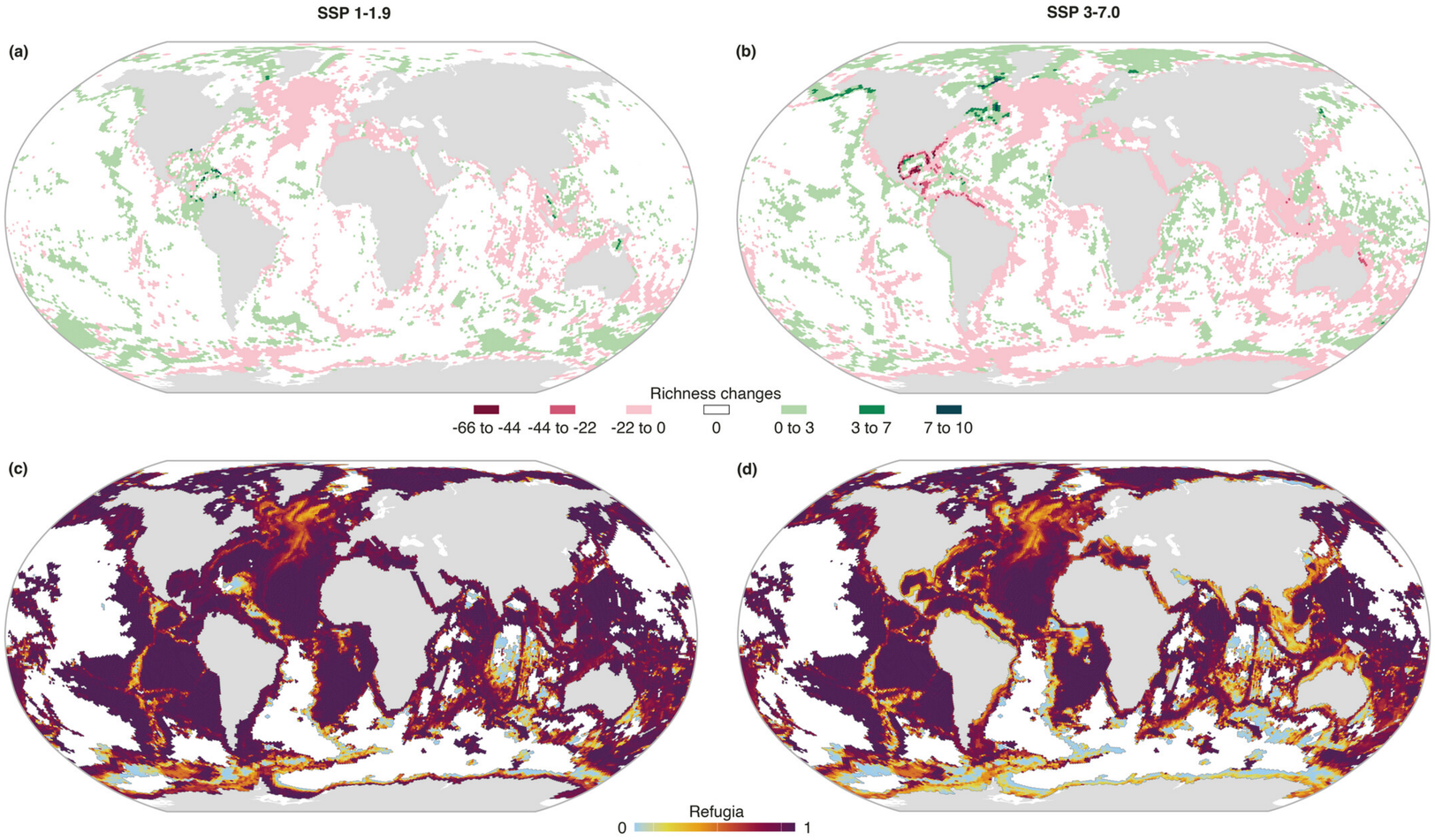
Goal: Delineate the global biogeography of CWCs and assessed **how their biodiversity patterns may shift under future climate change scenarios**.

Methods: Ensemble of machine-learning models predicted the distributions of 741 CWC species under present-day conditions and forecasted changes in species richness, community composition, and climate refugia under SSP1-1.9 and SSP3-7.0.


(a)
Model performance
1
0.75
0.5
0.25
0
AUC
Boyce (rescalled 0-1)
Sensitivity
AdaBoost
BRT
XgBoost
Cross validation models
(b)
AdaBoost
BRT
XgBoost
Ensemble
Ensemble const.
Predictive models
(c)
Relative contribution (%)
75
50
25
5
0
Temp. Min.
Temp. Max.
Rugosity
Slope
pH
Oxygen
Productivity
Currents
Salinity
Predictor variable



Major biodiversity hotspots in the Gulf of Mexico and the Caribbean Sea, and delineated **ten distinct bioregions**, each with varying species richness, depth distribution patterns, and generally low levels of endemicity.



Forecasts showed **poleward shifts in species distributions**, particularly under high emission scenarios, resulting in **biodiversity losses**, highlighting the need for urgent climate action, aligned with the Paris Agreement. Also global and **widespread conservation opportunities (e.g., refugia)**.

npj | biodiversity

Article

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Oceanographic connectivity strongly restricts future range expansions of critical marine forest species

Check for updates

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Climate change is driving a global redistribution of marine biodiversity. As habitats shift, oceanographic connectivity (the transport of dispersive stages via ocean currents) becomes a critical, yet poorly understood, factor that can either facilitate or hinder species' expansions into new areas. To quantify this influence, we developed a framework linking species distribution models with biophysical connectivity models to examine the redistribution of 467 marine forest species (seagrasses and brown macroalgae) under end-of-century climate change scenarios. Our projections show substantial habitat loss for both groups, reaching up to 50% (seagrasses) and 58% (brown macroalgae) of current habitats under higher emissions. Despite potential poleward expansions, oceanographic connectivity emerges as a major limiting factor. Accounting for average dispersal duration, range expansions are reduced by up to 38% in area (seagrasses) and 48% (macroalgae), and by up to 64% and 72% in distance, respectively. This reduction significantly increases the percentage of species facing a net loss of suitable habitat. Notably, well-defined dispersal barriers restrict expansions into highly suitable regions (e.g., the Okhotsk Sea, New Zealand, and the Arctic). Our findings underscore the need to explicitly integrate both habitat suitability and oceanographic connectivity to accurately predict marine biodiversity redistribution and inform effective conservation strategies.

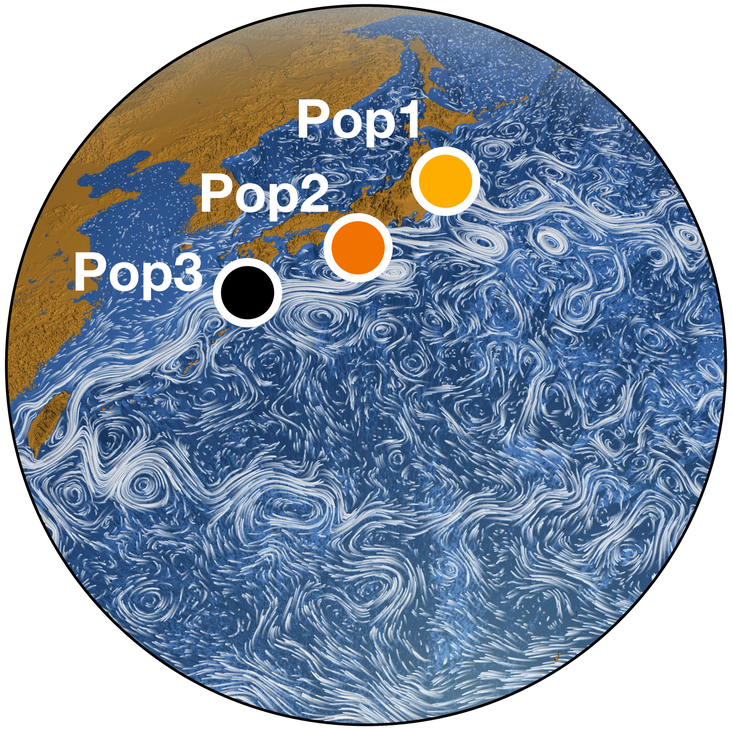
Suitable conditions

Safeguarded population

Where populations are allowed to persist locally, there are numerous opportunities for long-term conservation of species under climate change.

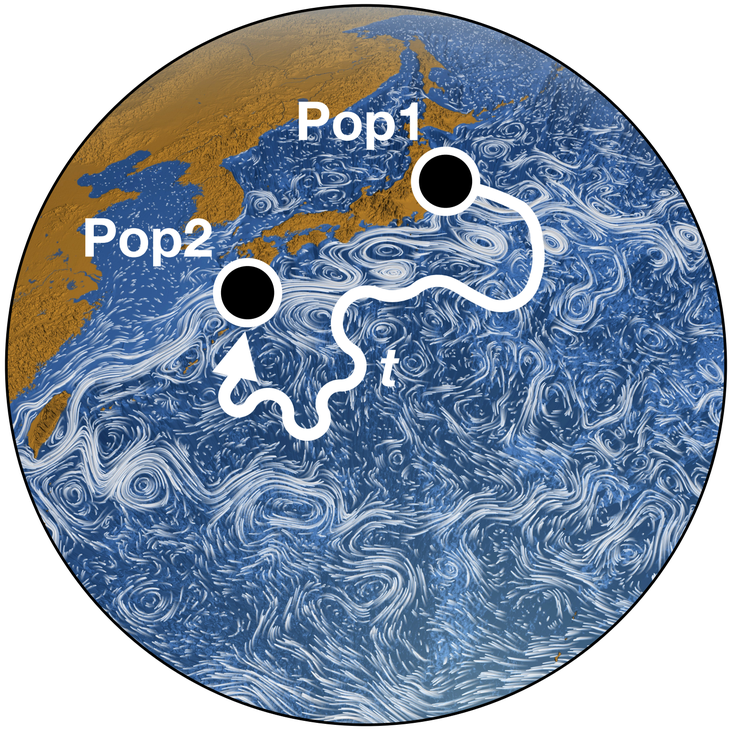
But, **where climate threatens populations**, their persistence relies on their ability to **move and track environmental changes**.

In the near future populations need to disperse to suitable areas.



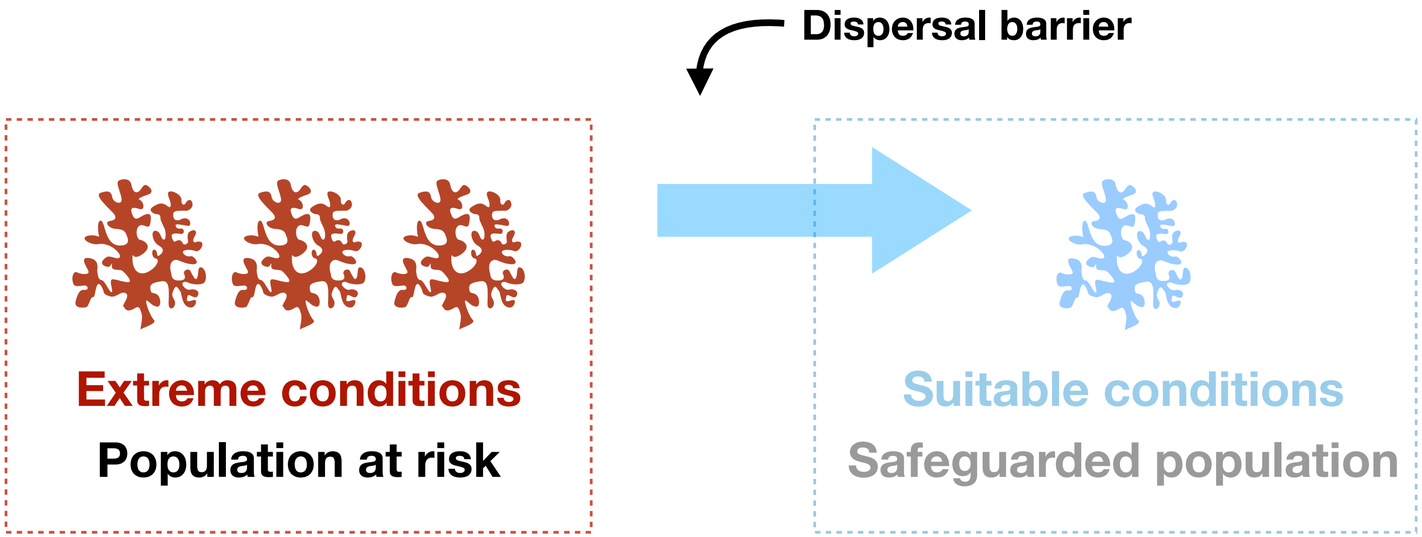
Historically, marine ecosystems were considered open, with populations moving freely over large spatial scales.

Studies systematically show species made of distinct populations, even for migratory species. **Implies the presence of barriers to dispersal**, i.e., to the movement of individuals between populations.



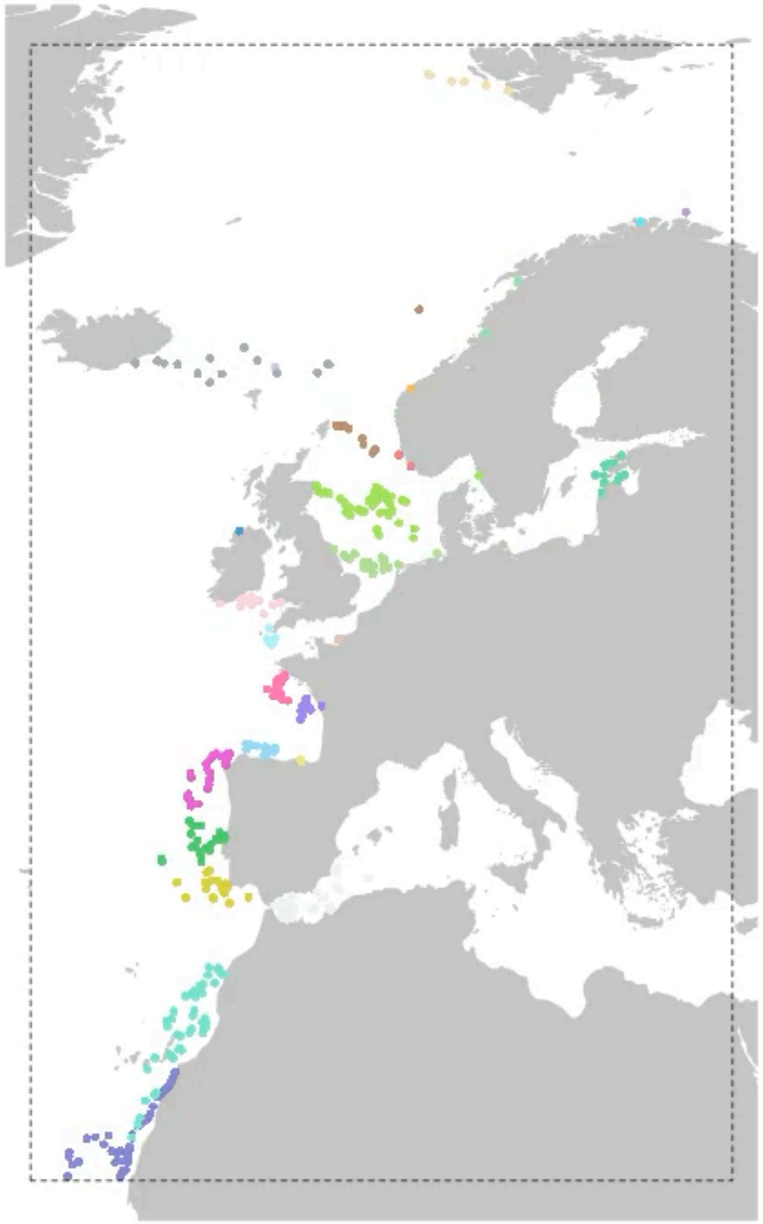
Population movement is shaped by the patterns of **ocean currents and the time a species is able to live in the water column** (e.g., larvae).

Favorable ocean currents between populations / species with high dispersal potential allows connectivity, while **oceanographic barriers / low dispersal potential can lead to strong population isolation.**



While species track suitable conditions, **ocean currents** may shape **dispersal barriers**, blocking movement, leaving them at risk.

What is the actual impact of oceanographic barriers on species shifting ranges under climate change?

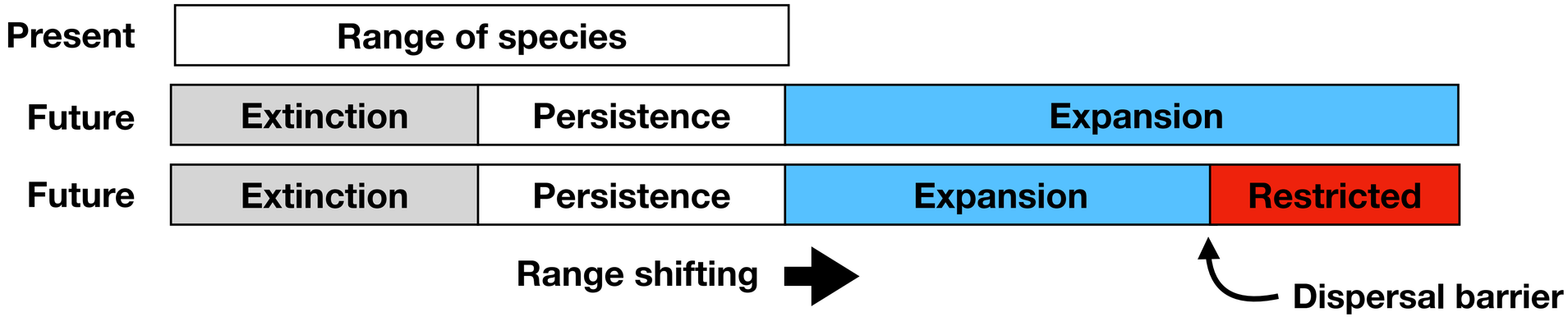


Biophysical modeling estimate population connectivity.

Virtual particles (e.g., fish larvae) **follow the direction and intensity of ocean currents in a virtual ocean**.

Can run globally, using **decades** of **high-resolution ocean current data** (e.g., Copernicus).

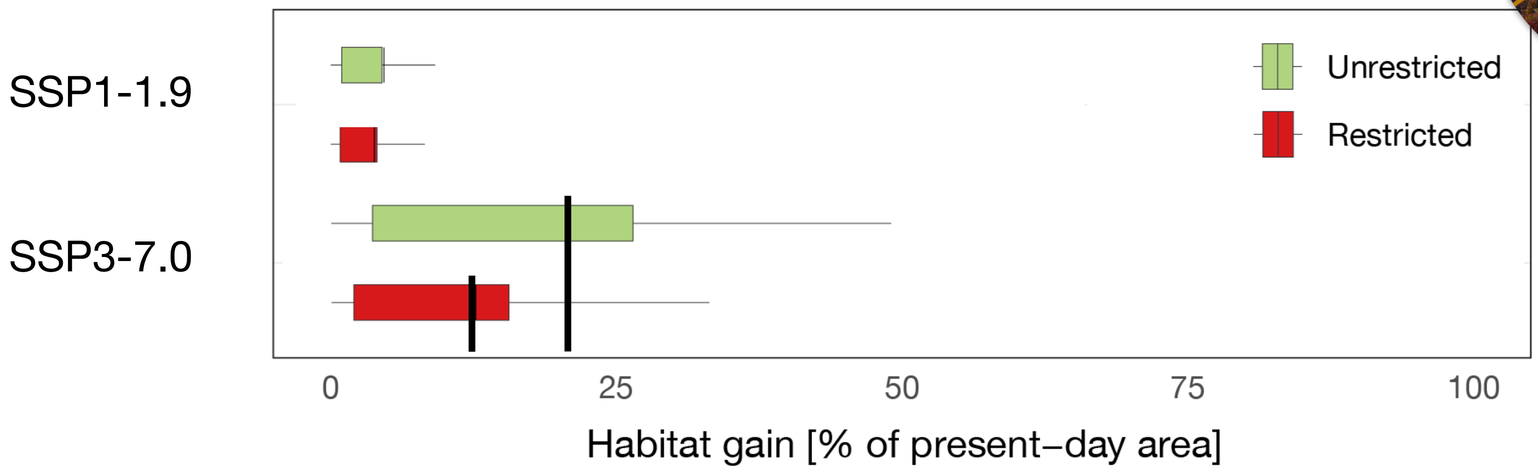
Connectivity events between locations are recorded.



Connectivity estimates between locations can be integrated in distribution models projecting the distribution of species.

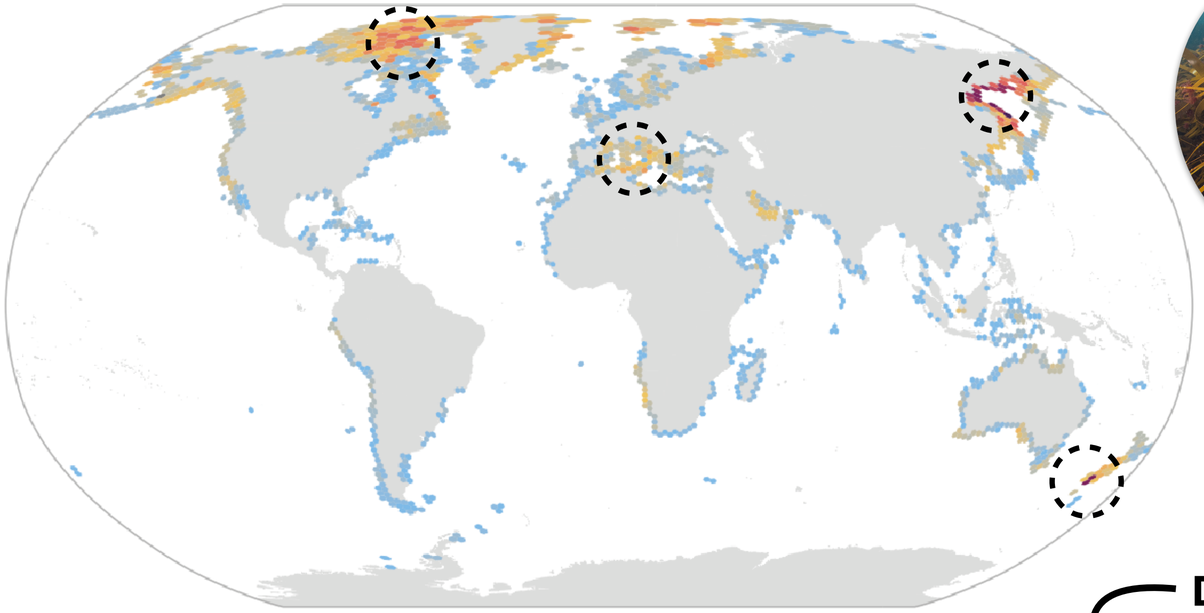
To **test the effect of dispersal barriers**, comparisons of distributions between the present and the future considering:

(1) **unlimited range expansions to suitable areas;** (2) **restricted range expansions due to oceanographic barriers**.



Under high emissions, substantial range expansions (>20% of their present distribution). **Considering dispersal barriers restricted the future colonization of new habitats by ~55%**, i.e., half of habitats are not colonized.

Implying that numerous populations might not be safeguarded.



Dispersal barrier

Extinction	Persistence	Expansion	Restricted

The **effect of these barriers can leave numerous regions uncolonized**, like in the NW Pacific, New Zealand, the Arctic and the central Mediterranean Sea, **increasing the negative impacts of climate change**.